Lecture 1: Random variables and probability

COMP 521 • Probability, Optimization, and Scientific Discovery • How to use these notes; Sections 1.1–1.5

> This handout is one part of the full COMP 521 notes. Section, equation, figure and listing numbers match the full notes, and references to anything outside this handout point there. The notation table is in "How to use these notes" below.

How to use these notes

Audience and prerequisites. You should be comfortable with multivariable calculus (partial derivatives, the chain rule, Taylor expansion), linear algebra (matrix products, eigenvalues, solving linear systems), and Python with NumPy. You do not need prior coursework in statistics or optimization; Section 1 rebuilds the probability you need. Research students who already know that material can skim Section 1 and start at Section 1.6.

Glass box, then black box. Each algorithm appears first as a few lines of JAX that you could have written yourself: Levenberg–Marquardt, Hamiltonian Monte Carlo, simulation-based calibration, the adjoint method. Only after that do we point to the production library that does the same job more robustly. A library you can reproduce is a library you can debug.

Boxes. Four kinds of box recur.

Key idea

The conceptual point a section exists to make.

Common pitfall

A mistake that experienced researchers make, usually with confidence.

In Jaxstro

How the idea appears in the research codes: Stellax (differentiable stellar evolution), Gravax (differentiable N-body dynamics), Startrax (fast stellar tracks), Progenax (stellar-population initial conditions), and the observation and inference layers Fluxax and Informax. The code listings in these notes never call those packages: they run with plain JAX so that you can execute every one of them. The boxes explain what changes when the toy is replaced by the real solver.

Check your understanding

A short question to answer before reading on. The answer follows inside the box; cover it and commit to your own answer first. Longer practice problems are in Appendix A, with solutions in Appendix B.

Running the code. Install JAX (`pip install "jax[cpu]"`); the listings were run with JAX 0.12 on a CPU. They enable 64-bit floats with `jax.config.update("jax_enable_x64", True)`. JAX

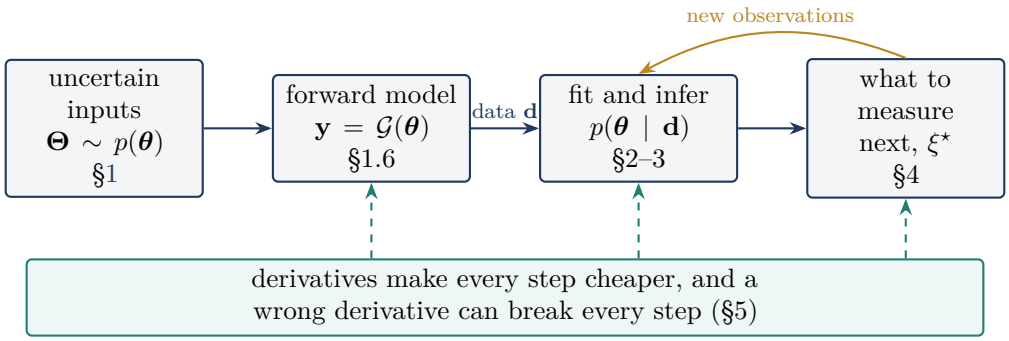
defaults to 32-bit floats, which are fine for training neural networks but too coarse for derivative checks and ill-conditioned linear algebra (Section 5.3).

Learning goals

After working through these notes you should be able to:

1. Distinguish a deterministic model, a random variable, a probability distribution, and an uncertain physical parameter, and say which kind of uncertainty (aleatoric, epistemic, numerical) a given error bar represents.
2. Read notation such as $\arg\min$, $\nabla\mathcal{L}$, $\mathbf{J}$, $\mathbf{H}$, $p(\boldsymbol{\theta} \mid \mathbf{d})$ and $\mathbb{E}[X]$ aloud and explain it in words.
3. Propagate uncertainty through a nonlinear model by Monte Carlo and by first- and second-order linearization, and predict when each fails.
4. Turn a noise model into a loss, derive the gradient-descent, Newton, Gauss–Newton and Levenberg–Marquardt updates, and explain the role of conditioning.
5. Write Bayes' theorem for a stellar-physics problem, compute a posterior analytically in the linear-Gaussian case, by the Laplace approximation, and by Hamiltonian Monte Carlo.
6. Check an inference code with simulation-based calibration and diagnose sampler output with effective sample size.
7. Distinguish sensitivity from identifiability and structural from practical identifiability; use the Fisher matrix in a unit-independent way.
8. Quantify the value of a proposed observation with expected information gain.
9. Verify a derivative; compute gradients through implicit solves with the adjoint method; and recognize the two places automatic differentiation is confidently wrong: discontinuities in an expectation and event times in a fixed-step solver.
10. Design an emulator evaluation that does not leak information between training and test sets.

Road map



Optimization appears in every box: fitting parameters, training an emulator, tuning a sampler, choosing a design. The scientific question decides *what* is optimized; that choice is the part no library makes for you.

Notation

Each symbol below keeps this meaning throughout. A few letters also carry their standard astrophysical meaning in one place (for example κ for opacity in Section 6, ρ for mass density); those uses are flagged where they appear.

Symbol	Read aloud	Meaning
X, x	"capital X", "x"	a random variable and one of its realizations
$\boldsymbol{\theta}$, $\boldsymbol{\Theta}$	"theta"	parameters of a physical or statistical model; $\boldsymbol{\Theta}$ when treated as random
p	"p"	number of parameters, $\boldsymbol{\theta} \in \mathbb{R}^p$ (bold $\mathbf{p}$ in Section 3.5 is an HMC momentum)
$\mathcal{G}(\boldsymbol{\theta})$	"G of theta"	the forward model, e.g. a Stellax track mapped to observables
$\mathbf{y}$	"y"	a model prediction, $\mathbf{y} = \mathcal{G}(\boldsymbol{\theta})$
$\mathbf{d}$	"d" (bold)	observed data
ξ	"xi"	an experimental design we choose
$\mathcal{L}(\boldsymbol{\theta})$	"loss of theta"	a scalar objective to be *minimized*
$p(\mathbf{d} \mid \boldsymbol{\theta})$	"d given theta"	likelihood; $\ell(\boldsymbol{\theta}) = \log p(\mathbf{d} \mid \boldsymbol{\theta})$
$p(\boldsymbol{\theta} \mid \mathbf{d})$	"theta given d"	posterior
$U(\boldsymbol{\theta})$	"U of theta"	negative log posterior, $-\ell(\boldsymbol{\theta}) - \log p(\boldsymbol{\theta})$ up to a constant (the potential energy in HMC)
$\mathcal{U}(\xi)$	"utility of xi"	the score of a design ξ, e.g. its expected information gain
$\nabla\mathcal{L}$	"grad loss"	gradient vector of a scalar function
$\mathbf{J} = \partial\mathbf{y}/\partial\boldsymbol{\theta}$	"Jacobian"	matrix of first derivatives of a vector function
$\mathbf{H} = \nabla^2\mathcal{L}$	"Hessian"	matrix of second derivatives of a scalar function
$\mathcal{I}(\boldsymbol{\theta})$	"Fisher information"	expected curvature of $-\ell$
$F_X(x)$	"F of x"	cumulative distribution function of X
$\boldsymbol{\Sigma}$, $\boldsymbol{\Sigma}_0$	"Sigma", "Sigma nought"	noise covariance; prior covariance
$\mathbb{E}[X]$, $\mathrm{Var}(X)$	"expectation", "variance"	probability-weighted mean; mean squared deviation
$\mathbb{H}[X]$	"entropy of X"	entropy
η	"eta"	step size or learning rate
ν	"nu"	Levenberg–Marquardt damping
λ	"lambda"	regularization strength; λ_i with an index is an eigenvalue; bold $\boldsymbol{\lambda}$ is the adjoint vector of Section 5.2
$\boldsymbol{\Lambda}$	"capital lambda"	population-level hyperparameters (IMF slope, binary fraction, age spread)
$\boldsymbol{\phi}$	"phi"	nuisance parameters
$\boldsymbol{\psi}$	"psi"	stellar or dynamical physics shared by all stars in a population
$\mathbf{z}_i$	"z i"	star-specific latent variables (true mass, mass ratio, …)
γ	"gamma"	slope of a power-law IMF, $\mathrm{d}N/\mathrm{d}m \propto m^{-\gamma}$
$\alpha(\boldsymbol{\Lambda}, \boldsymbol{\psi})$	"alpha of Lambda"	detection fraction: the probability that a star drawn from the population is detected (Section 3.6.1)
κ	"kappa"	condition number of a matrix (Section 2.4)
$\sim$, $\propto$	"distributed as", "proportional to"	declares a distribution; equality up to a constant factor

Vectors are bold, matrices are bold upright, and $^\mathsf{T}$ denotes a transpose. "log" is the natural logarithm, so information is measured in nats.

1 Random variables and probability

1.1 Deterministic physics, uncertain inputs

A Stellax run maps initial conditions and physics parameters to an evolutionary history,

$$\mathbf{y} = \mathcal{G}_{\text{Stellax}}(M, Z, t, \alpha_{\text{MLT}}, f_{\text{ov}}, \dots), \tag{1}$$

where M is the initial mass, Z the metallicity, t the age, α_{MLT} the mixing-length parameter, and f_{ov} an overshooting parameter. The output $\mathbf{y}$ might collect L, T_{eff}, radius, surface abundances, or the helium-core mass at the tip of the red-giant branch (TRGB). For fixed inputs and a fixed algorithm the output is fixed, up to numerical error.

Now pick a star at random from a cluster. You do not know its mass. Nothing about the physics has become random, yet the predicted luminosity is uncertain, because one of its inputs is. Probability is the bookkeeping for that situation.

Definition 1.1 (Random variable)**.** A *random variable* is a function $X : \Omega \to \mathbb{R}$ from a sample space Ω of possible outcomes to numbers.[1] A *realization* x is the value obtained in one outcome. A categorical outcome, such as remnant type, is a discrete random variable taking values in a finite set.

Key idea

Calling a quantity "random" is a statement about our information, not about the quantity. An uncertain mixing coefficient is modelled as a random variable because we do not know its value, not because it fluctuates from star to star.

1.2 Discrete probabilities, densities, and the CDF

A discrete random variable takes values in a countable set. A TOY remnant category might have $P(C = \text{WD}) = 0.70$, $P(C = \text{NS}) = 0.20$, $P(C = \text{BH}) = 0.10$: illustrative numbers that sum to one.

A continuous random variable such as mass has a *probability density function* (PDF) $p_M(m)$:

$$P(a \le M \le b) = \int_a^b p_M(m)\,\mathrm{d}m, \qquad \int_{-\infty}^{\infty} p_M(m)\,\mathrm{d}m = 1. \tag{2}$$

A density carries inverse units of its argument (here $M_\odot^{-1}$), so $p_M(1\,M_\odot)$ is *not* the probability that a star has exactly one solar mass; that probability is zero. Probability lives on intervals. The *cumulative distribution function* (CDF) is $F_M(m) = P(M \le m) = \int_{-\infty}^{m} p_M(u)\,\mathrm{d}u$; it rises monotonically from 0 to 1. A *quantile* inverts the CDF: the median $m_{1/2}$ satisfies $F_M(m_{1/2}) = 1/2$. The *mode* is where the density peaks. Mean, median and mode coincide for symmetric unimodal distributions (when the mean exists), and generally differ otherwise.

[1] Strictly, X must be measurable with respect to a probability measure on Ω; every function you will meet in these notes is.

1.2.1 Worked example: a single power-law IMF

Progenax draws initial masses from an initial mass function (IMF). The simplest teaching model is

$$\frac{\mathrm{d}N}{\mathrm{d}m} \propto m^{-\gamma}, \qquad m \in [m_{\mathrm{lo}}, m_{\mathrm{hi}}]. \tag{3}$$

$\mathrm{d}N/\mathrm{d}m$ counts stars per unit mass and is not normalized. (The astronomical literature often writes the IMF as $\xi(m)$; these notes reserve ξ for an experimental design.) The probability density is $p_M(m) = C\, m^{-\gamma}$ with

$$C = \frac{1-\gamma}{m_{\mathrm{hi}}^{1-\gamma} - m_{\mathrm{lo}}^{1-\gamma}} \quad (\gamma \neq 1), \qquad C = \frac{1}{\ln(m_{\mathrm{hi}}/m_{\mathrm{lo}})} \quad (\gamma = 1), \tag{4}$$

and CDF

$$F_M(m) = \frac{m^{1-\gamma} - m_{\mathrm{lo}}^{1-\gamma}}{m_{\mathrm{hi}}^{1-\gamma} - m_{\mathrm{lo}}^{1-\gamma}} \quad (\gamma \neq 1). \tag{5}$$

Salpeter's classic slope for massive stars is $\gamma \approx 2.35$ [1]. Realistic IMFs flatten below about $0.5\,M_\odot$ and are usually written as broken power laws or a lognormal joined to a power law [2, 3]; the single power law is here for its algebra.

Result 1.2 (Inverse-transform sampling)**.** If $U \sim \mathrm{Uniform}(0,1)$ and F is a continuous, strictly increasing CDF, then $M = F^{-1}(U)$ has CDF F. *Proof:* $P(M \le m) = P(F^{-1}(U) \le m) = P(U \le F(m)) = F(m)$, because F is increasing and U is uniform.

Solving Eq. (5) for m gives the sampler

$$M = F_M^{-1}(U;\gamma) = \left[m_{\mathrm{lo}}^{1-\gamma} + U\left(m_{\mathrm{hi}}^{1-\gamma} - m_{\mathrm{lo}}^{1-\gamma}\right)\right]^{1/(1-\gamma)}. \tag{6}$$

This produces a random population without making any stellar-evolution equation stochastic. It will also reappear in Section 5.5: because Eq. (6) is a smooth function of γ for fixed U, it lets us differentiate population statistics with respect to the IMF slope.

1.3 Expectation, variance, and standard error

The *expectation* of X is $\mathbb{E}[X] = \int x\, p_X(x)\, \mathrm{d}x$, and for any function g,

$$\mathbb{E}[g(X)] = \int g(x)\, p_X(x)\, \mathrm{d}x \tag{7}$$

(the "law of the unconscious statistician": you never need the density of $g(X)$ to average it). The *variance* is $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2$, and the *standard deviation* is $\sigma_X = \sqrt{\mathrm{Var}\,X}$, in the units of X.

A *standard error* is a different object: the standard deviation of an *estimator*. For independent, identically distributed draws $x_1, \ldots, x_N$ with variance σ^2, the sample mean $\bar{x} = N^{-1}\sum_i x_i$ has standard error $\sigma/\sqrt{N}$. This one formula governs every Monte Carlo estimate in these notes. Correlated draws, such as successive states of a Markov chain, break it; Section 3.5 shows how to repair it with an *effective* sample size.

1.4 Joint, conditional, and marginal distributions

Collect several uncertain inputs into a random vector, $\boldsymbol{\Theta} = (M, Z, \alpha_{\mathrm{MLT}}, f_{\mathrm{ov}}, f_{\mathrm{wind}})^{\mathsf{T}}$, where f_{wind} is a multiplier on a mass-loss prescription. A *joint* density $p(m, z)$ assigns density to combinations. The *conditional* density

$$p(m \mid z) = \frac{p(m, z)}{p(z)} \qquad (p(z) > 0) \tag{8}$$

describes mass once metallicity is known. *Marginalizing* integrates out what you do not keep: $p(m) = \int p(m, z)\, \mathrm{d}z$. Rearranging gives the product rule $p(m, z) = p(m \mid z)\, p(z)$, from which Bayes' theorem (Section 3) follows in one line.

Independence, $p(m, z) = p(m)p(z)$, is a strong claim. It does not follow from two parameters having different names. Uncertainties in opacity and in mixing efficiency are conceptually distinct, but if both were calibrated against the same solar model their *errors* are coupled.

The *covariance* is $\mathrm{Cov}(X_1, X_2) = \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)]$ and the correlation coefficient is $\rho_{12} = \mathrm{Cov}(X_1, X_2)/(\sigma_1\sigma_2) \in [-1, 1]$. Zero correlation does not imply independence: if $X \sim \mathcal{N}(0, 1)$ then X and X^2 are uncorrelated but completely dependent. For a linear combination $Y = \mathbf{u}^{\mathsf{T}}\boldsymbol{X}$ with covariance matrix $\boldsymbol{\Sigma}_X$,

$$\mathrm{Var}(Y) = \mathbf{u}^{\mathsf{T}}\boldsymbol{\Sigma}_X\mathbf{u}, \qquad \text{e.g. } \mathrm{Var}(aX_1 + bX_2) = a^2\sigma_1^2 + b^2\sigma_2^2 + 2ab\ \mathrm{Cov}(X_1, X_2). \tag{9}$$

The cross term is where compensating stellar prescriptions hide: a high opacity and a high mixing efficiency can push T_{eff} in opposite directions.

1.4.1 Changing variables in a density

If $Y = g(X)$ with g smooth and invertible, probability is conserved interval by interval, $p_Y(y)\,|\mathrm{d}y| = p_X(x)\,|\mathrm{d}x|$, so

$$p_Y(y) = p_X\big(g^{-1}(y)\big)\left|\frac{\mathrm{d}g^{-1}}{\mathrm{d}y}\right|, \qquad \text{and in } n \text{ dimensions} \qquad p_{\boldsymbol{Y}}(\boldsymbol{y}) = p_{\boldsymbol{X}}(\boldsymbol{x})\,\big|\det \partial\boldsymbol{x}/\partial\boldsymbol{y}\big|. \tag{10}$$

The Jacobian factor is the reason a density, unlike a probability, changes when you change units or coordinates. It explains why MAP estimates are not invariant under reparameterization (Section 3.2) and why log-parameters need a correction term when sampling (Section 3.5).

Worked example 1.3 (From an IMF to a luminosity function)**.** Take the IMF of Eq. (4) and the TOY relation $L = M^4$ (solar units). Then $m = L^{1/4}$ and $\mathrm{d}m/\mathrm{d}L = \frac{1}{4}L^{-3/4}$, so

$$p_L(L) = C\,(L^{1/4})^{-\gamma} \cdot \tfrac{1}{4}L^{-3/4} = \tfrac{C}{4}\,L^{-(\gamma+3)/4}.$$

A Salpeter IMF ($\gamma = 2.35$) becomes a luminosity function with slope -1.34. Forgetting the Jacobian factor would give slope -0.59: an error in the shape of the distribution, not just its normalization.

1.5 Three kinds of uncertainty, and one kind of error

Aleatoric uncertainty is variability of outcomes under a fixed conditioning set: draw a different star, get a different mass; repeat an exposure, get a different photon count. *Epistemic* uncertainty is missing knowledge: the true value of a reaction rate, a mixing coefficient, or the form of a missing

physical process. Both are represented with probability, but they behave differently: more stars shrink the uncertainty in an average over stars; more stars do not, by themselves, teach you a shared physics parameter (Section 1.7). Which label applies depends on what is conditioned on.

Numerical error is different in kind. Unconverged integration, a poorly placed mesh, a bug, or a wrong derivative is not a random variable with a known distribution. It is removed by convergence studies, conserved quantities, reference solutions, and derivative checks, not averaged away.

Source	Stellar example	Scientific response
Population variability	the mass of the next Progenax draw	sample an explicit population model
Measurement noise	photometric scatter	a defensible likelihood with its covariance
Unknown shared physics	the opacity or mixing law in Stellax	parameters with priors; calibration; tests against independent data
Model inadequacy	missing transport or mass-loss physics	improve the model, or add an explicit discrepancy term (Section 4.4)
Numerical error	unconverged solve; wrong event sensitivity	verify the computation before any inference

Common pitfall

A narrow posterior is not a correct posterior. If the noise model is wrong or a physical process is missing, more data make the error bars smaller around the wrong answer.

Exercises for this lecture

Exercises 2, 3, 5, numbered as in Appendix A of the full notes. Worked solutions are in Appendix B; attempt each one before reading its solution.

2. [P] Mean, median, mode. For the IMF $p(m) = Cm^{-2}$ on $[1, 10]\,M_\odot$: find C, $P(M > 5\,M_\odot)$, the median and the mean. Order the mode, median and mean, and explain the ordering.

3. [P] Log-uniform survives power laws. Masses are log-uniform on $[m_{\rm lo}, m_{\rm hi}]$ ($\gamma = 1$) and $L = M^{3.5}$. Derive $p_L(L)$ with Eq. (10). What general property of log-uniform distributions does the result illustrate?

5. [P] Signs of correlation. With $\sigma_1 = 2$, $\sigma_2 = 3$, $\rho_{12} = -0.5$, compute $\mathrm{Var}(X_1 + X_2)$ and $\mathrm{Var}(X_1 - X_2)$. Give a stellar-physics situation where two error sources are anticorrelated, and say which combination of them is then better known.

References

[1] E. E. Salpeter. The luminosity function and stellar evolution. *The Astrophysical Journal*, 121: 161, 1955.

[2] P. Kroupa. On the variation of the initial mass function. *Monthly Notices of the Royal Astronomical Society*, 322:231–246, 2001.

[3] G. Chabrier. Galactic stellar and substellar initial mass function. *Publications of the Astronomical Society of the Pacific*, 115:763–795, 2003.